



Cracking Password Assignment (Core)

Olubukola Omotayo
Coding Dojo Cybersecurity Boot Camp
17th November 2023



Password Cracking (Core)



Learning Objectives:

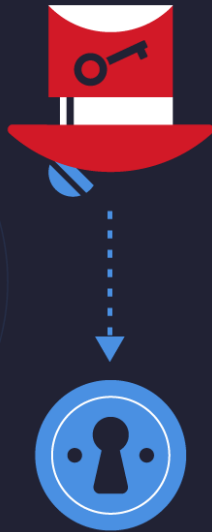
- Crack passwords using John the Ripper tool.



Overview

We are going to use a command line tool called John the Ripper. We are going to call the application, tell it what kind of hash it is, give it a file of hashes and that's all we need to give it for it to start showing the cracked passwords.

Reasons to Use John The Ripper



- Works with **Unix, Windows & Kerberos**
- Also compatible with **LDAP, MySQL & MD4** with the addition of extra modules
- Popular **password cracking tool**
- Preferred by **pentesters**
- Accessible on **multiple platforms**
- **Auto-detects** password hash types
- Can crack **multi-encrypted formats**



Instructions

Follow the instructions below to crack hashed passwords using John The Ripper:

1. Open Kali Linux.
2. [Download this file of hashes.](#) Save the file in your **Kali VM** as 'hashes.txt'
3. Open a terminal in Kali Linux.
4. Navigate to the folder in the terminal where you downloaded 'hashes.txt'
5. Choose the format NT and crack the hashes using the command below.

```
john -format=NT hashes.txtcopy
```

***Note: -format=NT is an option in John The Ripper that tells the program the hashes you downloaded are encrypted with NTLM. Since we're being explicit and telling the program what encryption is used, John The Ripper can attempt to crack these faster.**

Assignment Submission

Upload a single document with screenshots of John The Ripper cracking the passwords, and a list of the plaintext cracked passwords. Be sure to provide a sentence or two describing what each screenshot represents.

Assignment Solution:

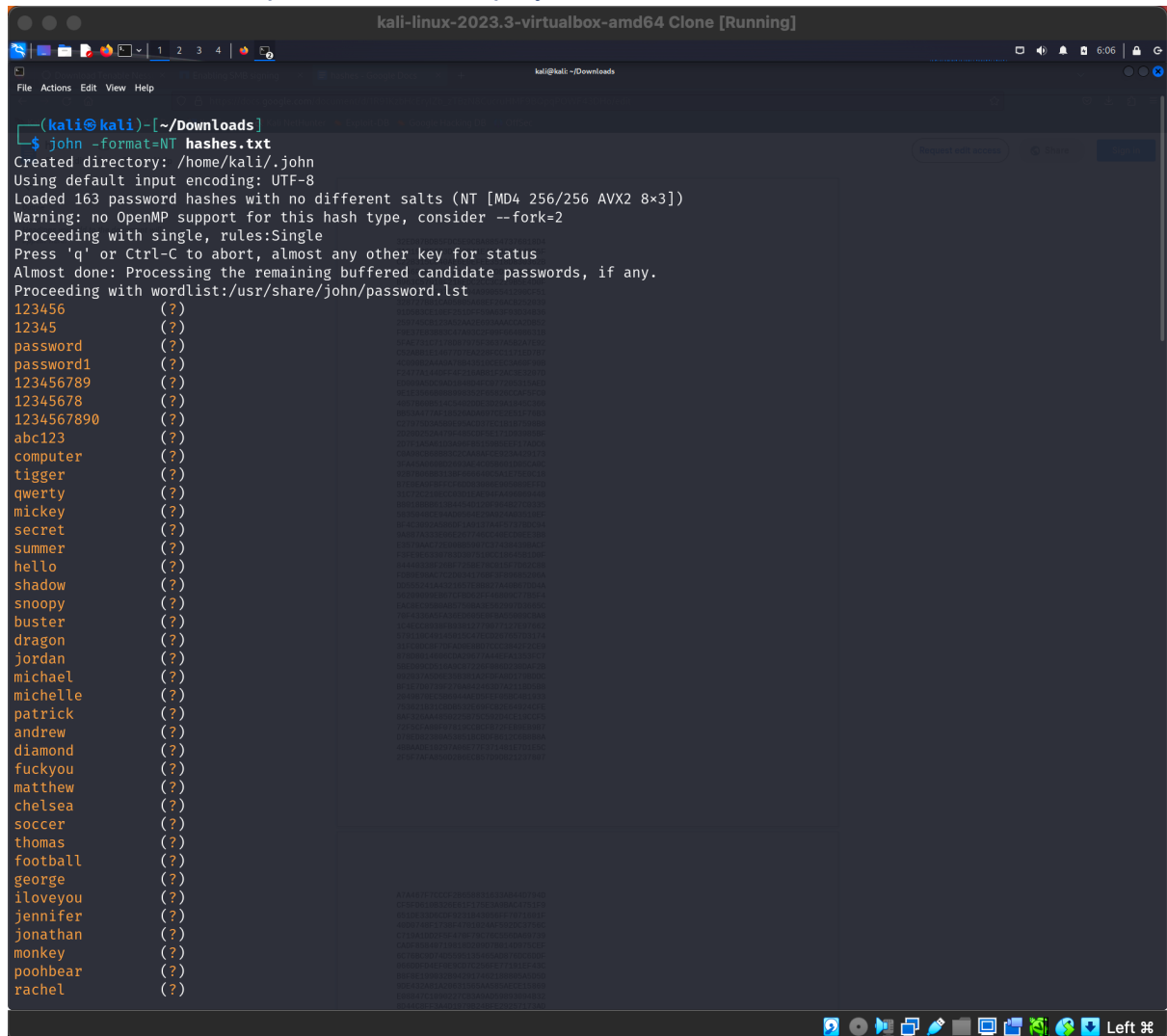
 hashes    

File Edit View Tools Help

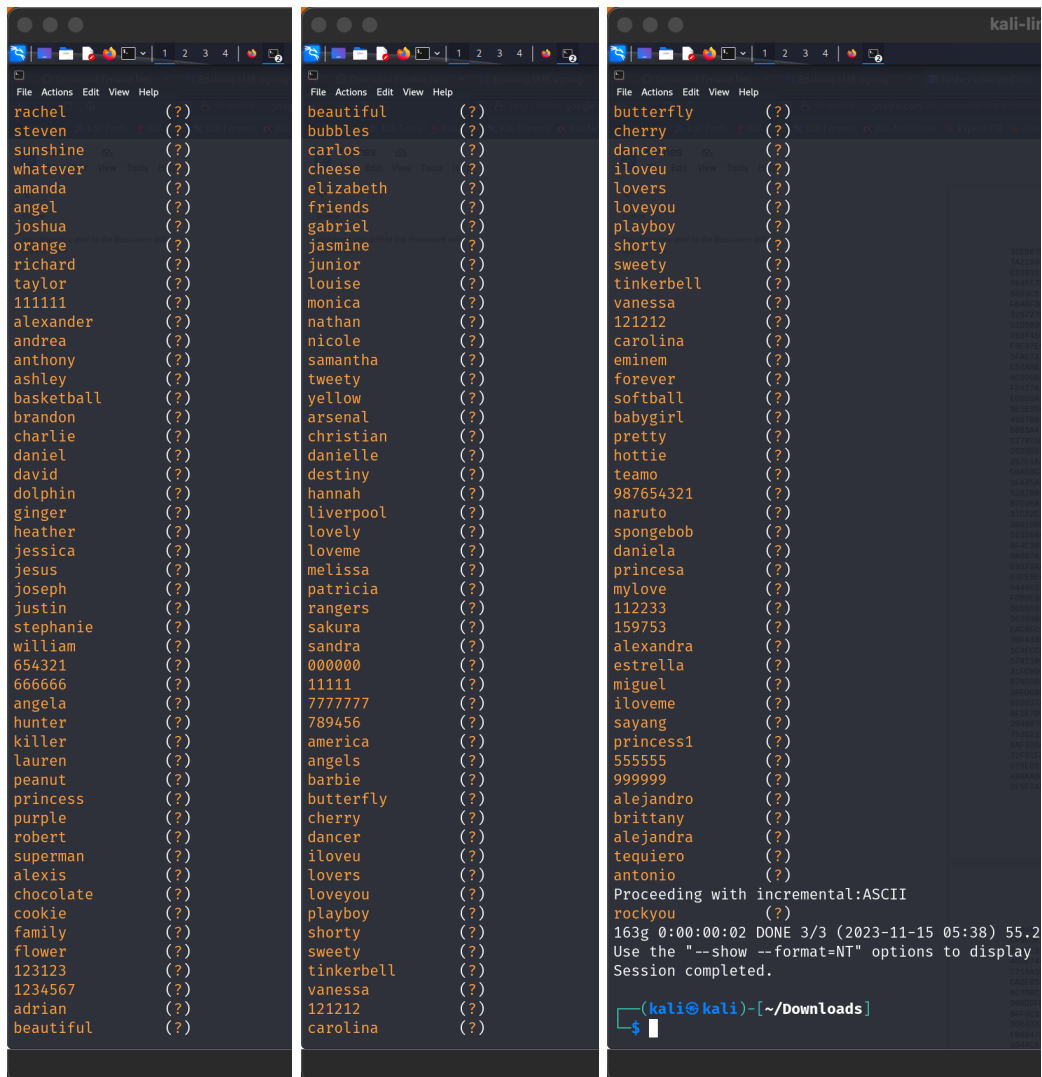


```
32ED87BDB5FDC5E9CBA88547376818D4
7A21990FCD3D759941E45C490F143D5F
C22B315C040AE6E0EFEE3518D830362B
8846F7EAAE8FB117AD06BDD830B7586C
B963C57010F218EDC2CC3C229B5E4D0F
FB4BF3DDF37CF6494A9905541290CF51
328727B81CA05805A68EF26ACB252039
91D5B3CE10EF251DFF59A63F93D34B36
259745CB123A52AA2E693AAACCA2DB52
F9E37E83B83C47A93C2F09F66408631B
5FAE731C7178D87975F3637A5B2A7E92
C52ABB1E14677D7EA228FCC1171ED7B7
4C090B2A4A9A78B43510CEEC3A60F90B
F2477A144DFF4F216AB81F2AC3E3207D
ED009A5DC9AD1848D4FC077205315AED
9E1E3566B088998352F65826CCAF5FC0
4057B60B514C5402DDE3D29A1845C366
BB53A477AF18526ADA697CE2E51F76B3
C27975D3A5B9E95ACD37EC1B1B7598B8
2D20D252A479F485CDF5E171D93985BF
2D7F1A5A61D3A96FB5159B5EEF17ADC6
C0A98CB68B83C2CAA8AFCE923A429173
3FA45A060BD2693AE4C05B601D05CA0C
92B7B06BB313BF666640C5A1E75E0C18
B7E0EA9FBFFCF6DD83086E905089EFFD
31C72C210ECC03D1EAE94FA496069448
B8018BBB613B4454D120F964B27C0335
5835048CE94AD0564E29A924A03510EF
BF4C3092A586DF1A9137A4F5737BDC94
9A887A333E06E267746CC40ECD0EE3B8
E3579AAC72E00BB5907C37438439BACF
F3FE9E6330783D307510CC18645B1D0F
.....
```

Using the 'John The Ripper' tool, passwords in the 'hashes.txt' file were cracked. The list of cracked passwords is displayed in the screenshot below:



```
kali-linux-2023.3-virtualbox-amd64 Clone [Running]
kali@kali: ~/Downloads
(kali@kali)~[~/Downloads]
$ john -format=NT hashes.txt
Created directory: /home/kali/.john
Using default input encoding: UTF-8
Loaded 163 password hashes with no different salts (NT [MD4 256/256 AVX2 8x3])
Warning: no OpenMP support for this hash type, consider --fork=2
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist: /usr/share/john/password.lst
123456 (?)
12345 (?)
password (?)
password1 (?)
123456789 (?)
12345678 (?)
1234567890 (?)
abc123 (?)
computer (?)
tiger (?)
qwerty (?)
mickey (?)
secret (?)
summer (?)
hello (?)
shadow (?)
snoopy (?)
buster (?)
dragon (?)
jordan (?)
michael (?)
michelle (?)
patrick (?)
andrew (?)
diamond (?)
fuckyou (?)
matthew (?)
chelsea (?)
soccer (?)
thomas (?)
football (?)
george (?)
iloveyou (?)
jennifer (?)
jonathan (?)
monkey (?)
poohbear (?)
rachel (?)
```



I observed that the complexity of these cracked passwords is really low. These passwords were simply:

• Lowercases	• Regular words
• Numbers only	• Short phrases
• Sequential numbers	• Names, etc.

This underscores the need to adhere to password complexity rules, in order to ensure that passwords are not easy to crack by 'John the Ripper' and similar tools.

Now I have a deeper understanding of the recommendation of using a combination of Uppercase, lowercase, numbers, special characters, non-sequential numbers, non-regular words, etc. to ensure password complexity.

This would secure passwords from being easily cracked by threat actors thereby securing the Confidentiality, Integrity and Availability of data.